Abhishek Nigam

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Summary

Software Engineer with 2.4+ years of experience as a Frontend Developer building scalable and responsive web applications. Holds a Master's degree (M.Tech CSE). Specialized in modern frontend (React.js, Next.js) with focus on performance, accessibility, and system design. Experienced in backend (Node.js, REST APIs) and workflow automation. Solved 350+ LeetCode problems and filed a U.S. patent for AI/ML-driven cloud optimization systems, applying data-driven approaches to improve infrastructure efficiency and product reliability.

Skills

Languages: JavaScript, TypeScript, C++, Java, SQL

Frontend: React.js, Next.js, React Query, Redux, Tailwind CSS, Webpack

Backend & Databases: Node.js, REST APIs, API Integration, MongoDB, MySQL

AI/ML: Generative AI, LLMs, RAG, Prompt Engineering, Ollama

Cloud & Tools: Git, GitHub, IBM Cloud, Chrome DevTools, Cypress, Postman, Jenkins, CI/CD Pipelines, Jira

Experience

Software Engineer — **Jan 2024 – Present**
IBM — Bengaluru

- Scaled platform throughput by enabling **4× faster VM restart recovery**, optimizing **Redux state management** and async API handling, supporting onboarding of **200K+ customers.**
- Implemented Redis-backed API caching with **React Query** for optimized data fetching and **client-side rendering (CSR),** reducing UI reload **latency by 50% (4s to 2s)** and improving daily active usage by **15%.**
- Built **RBAC-based conditional rendering** for IBM Power11 using state-driven UI**,** reducing user-facing downtime by **40% vs Power10.**
- Reduced disk-bound latency by leveraging **virtual persistent memory as an in-memory caching layer**, combined with **lazy loading and pagination**, improving initial load time for large VM datasets by **35**%.
- Automated UI workflows using **Cypress** with API validation and CI/CD**,** achieving end-to-end coverage and **reducing defects by 30%.**
- Resolved **20+ production** issues and delivered new features, improving **system reliability and reducing user-reported incidents.**

AI/ML Intern — **Jul 2023 – Dec 2023**
IIIT Delhi — New Delhi

- Introduced an **AI-powered recipe generation** system using **GPT-2** with **prompt engineering** for cuisine-based personalization. (Demo)
- Fine-tuned GPT-2 using **input-conditioned prompts** (ingredients + cuisines) and optimized generation parameters to reduce repetition, **improving relevance by 60%** and **reducing manual effort by 70%.**

Projects

Collab Canvas | React.js, TypeScript, Node.js, Context APIs, WebSockets

- Built a **real-time collaborative whiteboard** supporting **10+ concurrent** users with **room-based session isolation.**
- Designed an event-driven WebSocket architecture processing **1000+ drawing events per session**, enabling **live stroke synchronization, late-join replay,** and **mobile drawing via Pointer Events**. (Live | GitHub)

Infra Optimizer | React.js, JavaScript, MySQL, MongoDB, REST APIs

- Developed an **AI-driven dashboard** analyzing **VM usage metrics** to detect **underutilized resources** and provide infrastructure insights.
- Implemented **Slack webhook** alerting and approval workflows**,** integrating the tool into a **Jenkins CI/CD pipeline** for scheduled execution, reducing incident response time by **10 minutes** and improving cloud resource visibility.

Education

Indraprastha Institute of Information Technology, Delhi — **Jul 2022 – Jun 2024**
M.Tech in Computer Science and Engineering — CGPA: 8.62

Reva University, Bengaluru — **Jul 2017 – Jun 2021**
B.Tech in Computer Science and Engineering — CGPA: 9.59

Achievements

- Filed a **U.S. patent** on "**AI-Driven Adaptive VM Placement"**, proposing **30-40% cloud resource optimization**. (GitHub)
- Certified in **Microsoft Azure Data Fundamentals** (DP-900) and **Google Analytics**. (Link)
- Secured **AIR 2075** in GATE CSE 2022 and cleared **GATE CSE in 2021 and 2022.** (Link)